STANDATA interpretation 23-BCI-018

Building

Guard requirements for glazing in a window

Date Issued: January 2026

ON HOLD - UNDER REVIEW

Purpose

To clarify the application of guard requirements when glazing in a window is positioned below the minimum height requirements within the National Building Code – 2023 Alberta Edition (NBC(AE)).

Discussion

Guards are protective barriers that prevent accidental falls where a fall hazard exists in a building. These hazards may include open sides of stairs, landings, balconies, mezzanines, galleries, raised walkways, or fixed and operable windows. When there is an elevation difference of 600 mm or more between two floors, or between a floor and the next lower surface, the NBC(AE) generally requires a guard to reduce the risk of injury.

Designers may choose to use glazing in guards. It is important to acknowledge when glazing is used in a guard, it introduces two hazards associated with both falling and human impact. The NBC(AE) addresses these hazards separately.

The NBC(AE) requires a guard where there is no wall and there is an elevation difference of 600 mm or more between a floor and the next lower surface. When a window is located within a wall, the window is considered part of that wall assembly. If a window is installed in a wall where there is an elevation difference between surfaces of 600mm or more, and the bottom sill is below the minimum height prescribed in the NBC(AE), the window must either be designed as a guard or be protected with a guard.

If a person impacts or collides with a fixed or operable window located below the minimum height prescribed in the NBC(AE), and the glazing is not properly reinforced, the glass may shatter and cause injury. To limit this, the NBC(AE) requires the glazing to be safety glazing conforming to CAN/CGSB-12.1, “Safety Glazing,” or wired glass conforming to CAN/CGSB-12.11-M, “Wired Safety Glass.” In addition, if the window is openable, the NBC(AE) has provisions to limit a person from falling out the window.

Where a window is openable and below the minimum height prescribed in the NBC(AE), it may create a risk of a person falling out of the window when it is in the open position. Depending on the type of building and whether it is designed in compliance to Part 3 or Part 9 of the NBC(AE)), the NBC(AE) has provisions to limit any clear unobstructed opening to not more than 100 mm measured either vertically or horizontally.

Limiting the window opening to 100 mm reduces the risk of falling but does not prevent injuries from impact with the glazing. Therefore, compliance with safety glazing requirements, as specified in the NBC(AE), remains necessary to reduce the likelihood of injury resulting from accidental impact.

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Further clarification on the use and acceptance of WOCDs is provided in STANDATA Building Code Variance 23-BCV-001 Window opening control devices (WOCD).

Code References

Division A, Article 1.4.1.2.

1.4.1.2. Defined Terms
Guard means a protective barrier around openings in floors or at the open sides of stairs, landings, balconies, *mezzanines*, galleries, raised *walkways* or other locations to prevent accidental falls from one level to another. Such a barrier may or may not have openings through it.

Division C, Sentence 3.3.1.18.(1)

3.3.1.18. Guards
1) Except as provided in Sentence (5) and Article 3.3.2.9., a *guard* not less than 1 070 mm high shall be provided
- a) around any roof to which access is provided for purposes other than maintenance,
- b) at openings into smoke shafts referred to in Subsection 3.2.6. that are less than 1 070 mm above the floor, and
- c) at each raised floor, *mezzanine*, balcony, gallery, interior or exterior vehicular ramp, and at other locations where (see Note A-9.8.8.1.)
  - i) the difference in elevation is more than 600mm between the walking surface and the adjacent surface, or
  - ii) the adjacent surface within 1.2 m of the walking surface has a slope of more than 1 in 2.

Sentence 3.3.1.20.(8)

3.3.1.20. Transparent Doors and Panels
8) A window in a public area that extends to less than 1 000 mm above the floor and is located above the second *storey* in a *building* of *residential occupancy*, shall be protected by a barrier or railing to not less than 1 070 mm above the floor, or the window shall be non-openable and designed to withstand the lateral design loads for balcony *guards* required by Article 4.1.5.14.

Article 3.3.2.17.

3.3.2.17. Safety Glazing
1) Except as permitted in Sentence (3), glazing in all fixed and operable panels of doors shall conform to Class A of CAN/CGSB-12.1, “Safety Glazing.”
2) Except as permitted in Sentence (4), glazing in all fixed and operable panels of windows shall conform to Class A of CAN/CGSB-12.1, “Safety Glazing.”

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Article 3.3.4.8.

3.3.4.8. Protection of Openable Windows
1) Except as provided in Sentence (2), openable windows in *suites* of *residential occupancy* shall be protected by
a) a *guard* with a minimum height of 1 070 mm constructed in accordance with Article 3.3.1.18., or
b) a mechanism that can only be released with the use of tools or special knowledge to control the free swinging or sliding operation of the openable part of the window so as to limit any clear unobstructed opening to not more than 100 mm measured either vertically or horizontally.

Sentences 9.8.8.1.(1) and (4)

9.8.8.1. Required Guards (See Note A-9.8.8.1.)
1) Except as provided in Sentence (2) and except at the leading edge at the top of a *flight*, every surface to which access is provided, including but not limited to *flights* of steps and *ramps*, exterior landings, porches, balconies, *mezzanines*, galleries and raised *walkways*, shall be protected by a *guard* on each side that is not protected by a wall for the length where the difference in elevation is more than 600 mm between the walking surface and the adjacent surface within 1.2m.
4) Except as provided in Sentence (5), openable windows in *buildings* of *residential occupancy* shall be protected by
a) a *guard*, or
b) a mechanism that can only be released with the use of tools or special knowledge to control the free swinging or sliding operation of the openable part of the window so as to limit any clear unobstructed opening to not more than 100 mm measured either vertically or horizontally.
(See Note A-9.8.8.1.(4).)

A-9.8.8.1. Required Guards. The requirements relating to guards stated in Part 9 are based on the premise that, wherever there is a difference in elevation of 600 mm or more between two floors, or between a floor or other surface to which access is provided for other than maintenance purposes and the next lower surface, the risk of injury in a fall from the higher surface is sufficient to warrant the installation of some kind of barrier to reduce the chances of such a fall. A wall along the edge of the higher surface will obviously prevent such a fall, provided the wall is sufficiently strong that a person cannot fall through it. Where there is no wall, a guard must be installed. Because guards clearly provide less protection than walls, additional requirements apply to guards to ensure that a minimum level of protection is provided. These relate to the characteristics described in Notes A-9.8.8.3., A-9.8.8.5.(1) and (3), A-9.8.8.5.(4) and A-9.8.8.6.(1).

Examples of such surfaces where the difference in elevation could exceed 600 mm and consequently where guards would be required include, but are not limited to, landings, porches, balconies, mezzanines, galleries, and raised walkways. Especially in exterior settings, surfaces adjacent to walking surfaces, stairs or ramps often are not parallel to the walking surface or the surface of the treads or ramps. Consequently, the walking surface, stair or ramp may need protection in some locations but not in others. (See Figure A-9.8.8.1.) In some instances, grades are artificially raised close to walking surfaces, stairs or ramps to avoid installing guards. This provides little or no protection for the users. That is why the requirements specify differences in elevation not only immediately adjacent to the construction but also for a distance of 1.2 m from it. (See Figure A-9.8.8.1.)

Article 9.8.8.7.

9.8.8.7. Glass in Guards
1) Glass in *guards* shall be
a) safety glazing of the laminated or tempered type conforming to CAN/CGSB-12.1, “Safety Glazing,” or
b) wired glass conforming to CAN/CGSB-12.11-M, “Wired Safety Glass.”

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Application

This interpretation applies to windows located in buildings where glazing is positioned below the minimum height requirements in the NBC(AE) and may pose a risk of injury to persons.

Interpretation

1. Glazing panels that are fixed and positioned below the minimum height requirements of the NBC(AE) must:

   - address the risk of injury due to fall hazards associated with glazing by:
     - incorporating safety glazing of laminated or tempered type conforming to CAN/CGSB-12.1, "Safety Glazing," or
     - incorporating wired glass conforming to CAN/CGSB-12.11-M, "Wired Safety Glass," and
   - address the risk of injury due to the collision hazards associated with glazing by:
     - incorporating safety glazing of laminated or tempered type conforming to CAN/CGSB-12.1, "Safety Glazing," or
     - incorporating wired glass conforming to CAN/CGSB-12.11-M, "Wired Safety Glass."

2. Glazing panels that are part of openable windows positioned below the minimum height requirements of the NBC(AE) must:

   - address the risk of injury due to fall hazards associated with glazing by:
     - incorporating a mechanism that can only be released with the use of tools or special knowledge to control the free swinging or sliding operation of the openable part of the window to limit any clear unobstructed opening to not more than 100 mm measured either vertically or horizontally, and
     - incorporating safety glazing of laminated or tempered type conforming to CAN/CGSB-12.1, "Safety Glazing," or
     - incorporating wired glass conforming to CAN/CGSB-12.11-M, "Wired Safety Glass," and
   - address the risk of injury due to the collision hazards associated with glazing by:
     - incorporating safety glazing of laminated or tempered type conforming to CAN/CGSB-12.1, "Safety Glazing," or
     - incorporating wired glass conforming to CAN/CGSB-12.11-M, "Wired Safety Glass."

This INTERPRETATION is applicable throughout the Province of Alberta.

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Unless stated otherwise, all Code references in this STANDATA are to Division B of the National Building Code – 2023 Alberta Edition

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